# Week 04: two kinds of work

## Programming task

Functions and practice.
Then the individual Midterm.
Practice and assessment stay separate.

## Research task

After the Midterm: Research 3.
No research due during Week 4.
Discuss in Week 5 · proposed 5%.

Use this guide with the same Week notebook; do not create a second set of weekly tasks. Email: submit.progmis@gmail.com.

# Programming practice

Functions, arguments and returned values
Predict → run → change → repair → explain

# Three core tasks: understand the Python

## Predict a return

Trace rectangle_area(3, 4). Separate definition from call.

## Change and reuse

Call with (3, 2). Add 1 to the returned value. Predict both.

## Repair the function

The function prints but returns None. Fix the returned value.

Open Week_04.ipynb → Session 2. Keep a prediction, result and explanation for each task.

# Core 1–2: predict, then change one thing

```
01 def rectangle_area(width, height):
02     return width * height
03
04 print("defined")
05 result = rectangle_area(3, 4)
06 print(result)
```

1. Separate defining the function from calling it.
2. Call (3, 4): 12; call (3, 2): 6.
3. Use the returned 6 in result + 1: 7.

Use the complete Core 1 and Core 2 starters in the notebook; explain each changed result.

# Core 3: find and repair one mistake

```
01  def rectangle_area(width, height):
02      print(width * height)  # Deliberate missing-return bug.
03
04  result = rectangle_area(3, 4)
05  print("caller received:", result)
06  # Repair the function and check both returned values.
```

1. Predict the printed area and the caller’s result.
2. Repair print-only behavior so the function returns.
3. Retest returned values for (3, 4) and (3, 2).

Edit the matching Core 3 cell. Keep the wrong result, your repair and a retest.

# Done: explain the concept, then save

| Step | What to do / keep |
| --- | --- |
| **Program** | Required function and returned dictionary |
| **Tests** | All four common cases with all three fields asserted |
| **README and ZIP** | Purpose, commands, expectation, limitation and AI disclosure |
| **Own explanation** | Be ready to trace a case in your announced individual slot |

After practice, explain arguments and return values. Stop at the Midterm start time.

# How the Midterm fits

Begin in Session 2; continue in Session 3
Implement, test, package and review

# Midterm: follow the existing notebook task

| Step | What to do / keep |
| --- | --- |
| **Open / implement** | Midterm begins → summarize_request(lines, reviewed). |
| **Required rule** | Unreviewed: hold; else >=20: team; else: standard. Return totals too. |
| **Four tests** | A: 4/20/hold; B: 3/9/standard; C: 4/20/team; D: 6/22/team. |
| **Package** | midterm.py + test_midterm.py + README.md; include AI disclosure. |
| **Keep / explain** | Fresh-folder test result, ZIP and backup; oral slot announced. |

Use the complete input table in the notebook. Threshold 16 and main/import detail stay optional.

# Research input validation

## What to research

Find out why programs check input before using or saving it.

## Possible directions

Choose one field and an invalid value. Consider the allowed values, the message to show and what should remain unchanged.

## A short summary example

A container count must be a positive integer. If the input is “many”, ask for a whole count and leave saved stock unchanged.

Discuss in Week 5 · summarize in your own words and cite your sources.